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Reader's Guide: How π Emerges from Binary Multiplication

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This guide explains, step by step, how the number $\pi = 3.14159\dots$ appears in a problem that has nothing to do with circles. It requires no mathematical background beyond knowing how to multiply.

1. Multiplying in Binary

Computers multiply numbers in binary (base 2), using only the digits 0 and 1. The same rules apply as in decimal — multiply each digit, add the columns, carry the overflow.

Here is $13 \times 11 = 143$ in binary:

1 1 0 1	(13 in binary)
× 1 0 1 1	(11 in binary)

1 1 0 1	13×1
1 1 0 1 .	13×1 , shifted left
0 0 0 0 . .	13×0 , shifted left twice
1 1 0 1 . . .	13×1 , shifted left three times

1 0 0 0 1 1 1 1	(143 in binary)

Each column is added up. When a column totals more than 1, the excess is *carried* to the next column — exactly like carrying in decimal addition.

2. The Carry Chain

Reading the columns from right to left, the column sums and carries for 13×11 are:

Column:	0	1	2	3	4	5	6
Sum:	1	1	1	3	1	1	1
Carry in:	0	0	0	0	1	1	1
Total:	1	1	1	3	2	2	2
Output bit:	1	1	1	1	0	0	0
Carry out:	0	0	0	1	1	1	1

The *carry chain* — the sequence 0, 0, 0, 0, 1, 1, 1, 1 — records the overflow at each step. At column 3, the sum exceeds 1 for the first time, and a carry is born. Once present, it persists.

The key insight from probability theory (Diaconis and Fulman, 2009): if the input digits are random, the carry at each position depends only on the *current* carry and the column sum — not on anything earlier. This makes the carry chain a **Markov process**, amenable to spectral analysis.

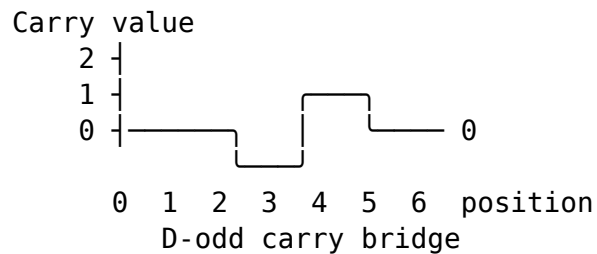
3. How Many Digits Does the Product Have?

When we multiply two 4-bit numbers (8 through 15), the product can have either 7 or 8 bits:

- $8 \times 8 = 64 \rightarrow 7$ bits (1000000)
- $11 \times 11 = 121 \rightarrow 7$ bits (1111001)
- $12 \times 12 = 144 \rightarrow 8$ bits (10010000)
- $15 \times 15 = 225 \rightarrow 8$ bits (11100001)

The boundary is at 128: products below 128 have 7 bits, products at or above 128 have 8 bits.

We call the 7-bit products **D-odd** (the product has an *odd* number of digits, $D = 2K - 1 = 7$). For D-odd products, the carry chain starts and ends at zero — a “bridge” that goes up and comes back down, like a ball thrown into the air.



4. The Four Sectors

Each 4-bit number has the form $1abc$ in binary. The **second-highest bit** a divides the numbers into two groups:

- **Low group** ($a = 0$): 8, 9, 10, 11 (binary 1000 through 1011)
- **High group** ($a = 1$): 12, 13, 14, 15 (binary 1100 through 1111)

Since we have two numbers X and Y , their second bits define four **sectors**:

	Y: low (8–11)	Y: high (12–15)
X: low (8–11)	Sector (0,0) 16 pairs 16 D-odd ✓	Sector (0,1) 16 pairs 8 D-odd ✓
X: high (12–15)	Sector (1,0) 16 pairs 8 D-odd ✓	Sector (1,1) 16 pairs 0 D-odd ✗

A proven fact (Theorem 4): Sector (1,1) has *zero* D-odd pairs. When both numbers are in the high group, the product always overflows to 8 bits. This is not a coincidence — it holds for all K , because if $X \geq 3 \cdot 2^{K-2}$ and $Y \geq 3 \cdot 2^{K-2}$ (both second bits are 1), then $XY \geq 9 \cdot 2^{2K-4} > 2^{2K-1}$, so the product always has $2K$ bits.

5. The Carry Weight and the Sector Ratio

For each D-odd pair, we read the carry chain and assign a **weight** of +1, 0, or -1. The “cascade” method scans the carry chain from top to bottom and reads the carry value c just below the highest nonzero carry position. The weight is $c - 1$: if $c = 0$, the weight is -1; if $c = 1$, the weight is 0; if $c \geq 2$, the weight is positive.

We add up these weights across all D-odd pairs in each sector: - σ_{00} = total weight from sector (0,0) - σ_{10} = total weight from sector (1,0)

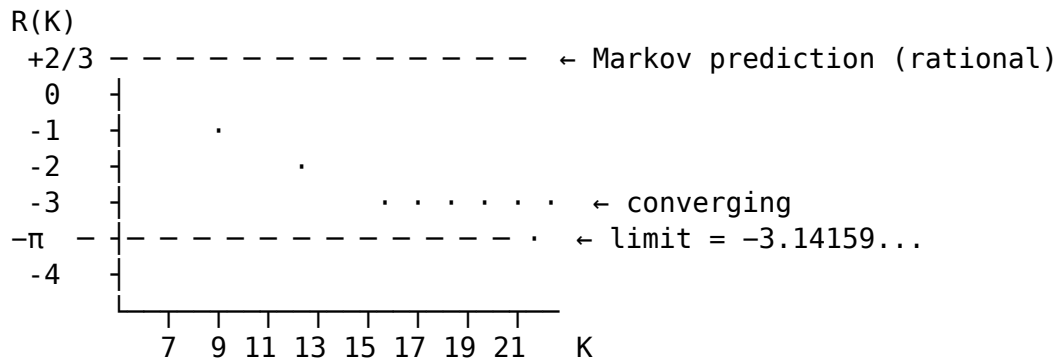
The **sector ratio** is their quotient:

$$R(K) = \frac{\sigma_{10}}{\sigma_{00}}$$

This is a purely combinatorial quantity: it counts carry patterns in binary multiplication, weighted by the cascade valuation. No geometry, no circles, no angles.

6. The Punchline: $R(K)$ Converges to $-\pi$

We computed $R(K)$ for every value of K from 4 to 21 by exhaustive enumeration — checking every single pair of K -bit numbers. At $K = 21$, this means examining over **one trillion** pairs.



K	Pairs checked	$R(K)$	Gap from $-\pi$
7	4,096	-0.09	3.05
11	~1 million	-1.55	1.59
15	~270 million	-2.82	0.32
19	~69 billion	-3.110	0.032
21	~1.1 trillion	-3.133	0.008
∞ (extrapolated)		-3.1416 ...	< 0.001

After Richardson extrapolation (a standard acceleration technique), the limit matches $-\pi = -3.14159 \dots$ to **4 significant digits**.

7. Why Is This Surprising?

Three things make this result remarkable:

1. The Markov model is completely wrong. If carries at each position were independent (the Markov assumption), the sector ratio would approach $R = +2/3$ — a positive rational number. The observed answer instead points toward $-\pi$ — negative, irrational, transcendental. The sign is wrong, the magnitude is wrong by a factor of 5, and the number itself is of a fundamentally different mathematical nature.

2. π has no reason to be here. The number π is the ratio of a circle's circumference to its diameter. Our problem involves no circles, no angles, no curves — just binary digits, column sums, and carries. The connection to π is deep: it enters through the Dirichlet character χ_4 , which satisfies $L(1, \chi_4) = \pi/4$ (the Leibniz series $1 - 1/3 + 1/5 - 1/7 + \dots = \pi/4$). The carry chain's spectral structure naturally selects this character.

3. A sharp phase transition is proposed. In the LMH model of [E], increasing the correlation between adjacent carries continuously transforms the spectral sum from $+2/3$ (Markov, subcritical) through 0 (critical point) to $-\pi$ (supercritical). The governing Möbius transformation is

$$S(A) = \frac{2(1 - A)}{3 - 2A}$$

where A measures the strength of digit correlations. Matching this model sum to $-\pi$ gives $A^* = (3\pi + 2)/(2(\pi + 1)) \approx 1.379$. If the physical sector ratio is governed by the LMH, this is the corresponding critical value.

8. What Is Proved and What Is Conjectured

Result	Status
The closed-form Möbius transformation $S(A) = 2(1 - A)/(3 - 2A)$	Proved (Theorem 1)
The Markov baseline $R_{\text{Markov}}(L) = +2/3 + O(L^{-2})$	Proved (Theorem 2)
The spectral zeta function reproduces $\zeta(2s)$	Proved (Theorem 3)
No D-odd pair has sector (1,1)	Proved (Theorem 4)
The count ratio N_{10}/N_{00} is a closed-form logarithmic constant	Proved (Theorem 5)
Cascade stopping depths in sector (0,0) obey structural rigidity	Proved (Theorem 6)
The first nonzero carry is exactly 1 $R(\infty) = -\pi$	Proved (Theorem 7) Conjectured (4.0-digit numerical evidence)

The conjecture reduces to proving two things: (i) a scalar constant equals -4 , and (ii) the spectral gap $1/2$ is preserved under the D-odd boundary condition. The companion paper [L] now realizes the same stopping-time channel as a canonical weighted first-return resolvent and shows that the current carry-side analytic object captures Euler/local-factor structure but does not yet transfer the L -function zeros. The algebraic framework is complete; the remaining step is analytical.

9. The Bigger Picture

The conjectural limit $R \rightarrow -\pi$ is one facet of a broader connection between binary carries and number theory:

- The **Riemann zeta function** $\zeta(s)$, which encodes the distribution of primes, is approximated by ensemble-averaged spectral determinants of carry companion matrices [B].
- The **spectral zeta function** of the Markov carry bridge reproduces $\zeta(2s)$ at all even arguments (Theorem 3 of this paper).
- Conjecturally, the sector ratio converges to $-4L(1, \chi_4) = -\pi$, connecting to the **Dirichlet L -function** of the character modulo 4.
- Together, $\zeta(s)$ and $L(s, \chi_4)$ form the **Dedekind zeta function** of the Gaussian integers $\mathbb{Z}[i]$: $\zeta_{\mathbb{Q}(i)}(s) = \zeta(s) \cdot L(s, \chi_4)$.

Binary multiplication, through its carry structure, already produces one factor of this Dedekind zeta function via the carry polynomial ensemble [B], and conjecturally contributes the second through the sector ratio studied here.

Glossary

- **Binary (base 2):** A number system using only digits 0 and 1.
 - **Carry:** The overflow when a column sum exceeds the base. In base 2, a column sum of 2 produces a carry of 1.
 - **D-odd:** A product whose binary representation has exactly $2K - 1$ digits (an odd number).
 - **Sector:** One of four groups determined by the second-highest bit of each factor.
 - **Sector ratio $R(K)$:** The ratio of carry-weighted sums in sectors (1,0) and (0,0).
 - **Markov process:** A random process where the next state depends only on the current state.
 - **Phase transition:** A qualitative change in behavior as a parameter crosses a threshold.
 - **Richardson extrapolation:** A numerical technique that accelerates convergence by eliminating leading error terms.
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